

Final review: key points

Chapt 1 Matrix Algebra

§1.4 Matrix Inverse

§1.5 Rank of Matrix

§1.6 Eigenvalue、Eigenvector & Trace of Matrix

§1.7 Positive Definite Matrix and Non-negative Definite Matrix

§1.8 Extremal Problem of Eigenvalue

- Quadratic minimum value theorem, Quadratic maximum value,
- Theorem (Extended) Cauchy–Schwarz Inequality,

Class Exercises and Assignments

- (4) If A is p dimensional symmetric matrix and its eigenvalues are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$, then

$$\max_{x \neq 0} x'Ax / x'x = \lambda_1 \quad (1.8.3)$$

$$\min_{x \neq 0} x'Ax / x'x = \lambda_p \quad (1.8.4)$$

The equality holds when $x = t_1$ and $x = t_p$, respectively.

- (5) If A is p dimensional symmetric matrix, B is p dimensional positive definite matrix and $\mu_1 \geq \mu_2 \geq \dots \geq \mu_p$ are the eigenvalues of $B^{-1}A$ and the corresponding eigenvectors $t_1, t_2, \dots, t_p$ satisfying $t_i' B t_j = 0, 1 \leq i \neq j \leq p$, then

$$\max_{x \neq 0} x'Ax / x'Bx = \mu_1 \quad (1.8.5)$$

$$\min_{x \neq 0} x'Ax / x'Bx = \mu_p \quad (1.8.6)$$

The equality holds when $x = t_1$ and $x = t_p$, respectively.

§1.9 Application

- See PPT for Chapter one (Review of Matrix Analysis)
- **Exercise 1.1.** Let $\Sigma > 0$ with order p , for any $p \times p$ symmetric matrix B , please prove that

$$\lambda_{\min}(\Sigma^{1/2} B \Sigma^{1/2}) \geq \lambda_{\min}(B) \lambda_{\min}(\Sigma)$$

Chapter 2: Random vector

§2.2 multivariate distribution, joint distribution, marginal distribution, conditional distribution.

§2.3 Numerical characteristics of random vectors

➤ Expectation, Covariance matrix (Σ) and their properties ,

Correlation Matrix (R) , The relationship between R and Σ .

$$R = V^{-1/2} \Sigma V^{-1/2}$$

with $V = \text{diag}\{\sigma_{11}, \sigma_{22}, \dots, \sigma_{pp}\}$.

§2.4 Mahalanobis Distance.

§2.5 Transformation of Random Vectors

Chapt3: Multinormal distribution

§3.1 Multinormal distribution

The probability density function (pdf)

$$f(\mathbf{x}) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right]$$

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \cdot \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}$$

§3.2 Properties of Multivariate Normal Distribution

$\mathbf{x}$ obeys multivariate normal distribution, if and only if any of its linear functions of $\mathbf{x}$ obey single normal distribution;

An arbitrary linear transformation of a multivariate normal vector is also a multivariate normal vector, but this conclusion does not hold for linear

combinations.

The conditional expectation and conditional covariance matrix of : x_1
given x_2

$$\begin{aligned}\mu_{1.2} &\equiv E(x_1 | x_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \\ \Sigma_{11.2} &\equiv Var(x_1 | x_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\end{aligned}$$

§3.3 Multiple and partial correlation coefficient

► **Multiple Correlation Coefficient** between y and x , denoted by $\rho_{y \cdot 1, 2, \dots, p}$, measures the degree of correlation between y and $x_1, x_2, \dots, x_p$

$$\rho_{y \cdot 1, 2, \dots, p} = \max_{l \neq 0} \rho(y, l'x) = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2}$$

when $l = \Sigma_{xx}^{-1} \sigma_{xy}$, the equality holds up

The largest correlation coefficient between y and the linear function of x is called the Multiple Correlation Coefficient, denoted by $\rho_{y \cdot x}$ or

$$\rho_{y \cdot 1, 2, \dots, p}$$

Know how to find the multiple correlation coefficient

Exercise 1. Assume $x = (x_1, x_2, x_3)'$, $x \sim N_3(\mu, \Sigma)$ with $\mu = (6, 3, 4)'$ and

$$\Sigma = \begin{bmatrix} 6 & -3 & -3 \\ -3 & 5 & 1 \\ -3 & 1 & 5 \end{bmatrix}$$

Then, the partial correlation coefficient of $x = (x_1, x_2)'$ given x_3 , $\rho_{12.3}$

=_____.

Exercise 2. In problem 1, the multiple coefficient of x_1 and (x_2, x_3) , $\rho_{1.23}$

=_____.

Exercise 1. $-\sqrt{14}/7$, Exercise 2. $\sqrt{2}/2$

§3.4 Maximum Likelihood Estimation (MLE) and its properties

*The formula of MLE: $\hat{\mu} = \bar{x}$, $\hat{\Sigma} = \frac{1}{n}A = \frac{n-1}{n}S$, where $A =$

$\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$ represents sample dispersion matrix.

Properties of estimators: $E(\bar{x}) = \mu$, $Var(\bar{x}) = \frac{1}{n}\Sigma$, $E(\hat{\Sigma}) = \frac{n-1}{n}\Sigma$.

§3.5 Sample Distribution about $\bar{x}$ and $(n-1)S$

$$\bar{X} \sim N_p(\mu, \frac{1}{n}\Sigma), \quad (n-1)S \sim W_p(n-1, \Sigma)$$

Wishart Distribution: A random matrix constructed from independent random vectors x_i with $x_i \sim N_p(0, \Sigma)$ follows a Wishart distribution with n degrees of freedom, denoted as $\sum_{i=1}^n x_i x_i' \sim W_p(n-1, \Sigma)$

Wishart distribution's properties:

► Properties of Wishart distribution:

1. (Additivity) If $W_i \sim W_p(n_i, \Sigma)$, $i = 1, 2, \dots, k$ are mutually independent, then,

$$\sum_{i=1}^k W_i \sim W_p(\sum_{i=1}^k n_i, \Sigma).$$

2. If a random matrix $W_{p \times p} \sim W_p(n, \Sigma)$, $C_{p \times p}$ is a non singular matrix, then

$$CWC' \sim W_p(n, C\Sigma C').$$

3. If $X_i \sim N_p(\mu, \Sigma)$, $i = 1, 2, \dots, n$, are mutually independent, then the sample dispersion matrix

$$(n-1)S = \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \sim W_p(n-1, \Sigma)$$

with $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Chapter 4: Multivariate Normal Population Statistical Inference

§4.2 Single Population Mean Inference:

Hotelling distribution: $x \sim N_p(0, \Sigma)$, $W \sim W_p(n, \Sigma)$, x and W are

mutually independent, then the distribution of $T^2 = nx'W^{-1}x$ is

Hotelling distribution with free degree n . We denote

$$T^2 = nx'W^{-1}x \sim T^2(p, n).$$

Properties:

1) Hotelling distribution T^2 is unrelated to Σ

➤ Hotelling distribution T^2 is unrelated to Σ

$$T^2 = n(\Sigma^{-1/2}x)'(\Sigma^{-1/2}W\Sigma^{-1/2})^{-1}(\Sigma^{-1/2}x)$$

➤ $\Sigma^{-1/2}x \sim N_p(0, I), \quad \Sigma^{-1/2}W\Sigma^{-1/2} \sim W_p(n, I).$

2) Hotelling distribution T^2 can be transformed into F distribution

$$\frac{n-p+1}{pn} T^2(p, n) \sim F(p, n-p+1)$$

Suppose $x_1, x_2, \dots, x_n$ is a sample of population $x \sim N_p(\mu, \Sigma)$, where

$$\Sigma > 0, n > p.$$

Definition of Hotelling statistic and its properties.

Consider the following hypotheses test:

$$H_0: \mu = \mu_0 \text{ vs } H_1: \mu \neq \mu_0.$$

When Σ is unknown, the Hotelling (霍特林) statistic (**To remember**)

$$T^2 = n(\bar{X} - \mu)S^{-1}(\bar{X} - \mu)'$$

is the sample squared Mahalanobis distance from the sample mean vector

to the population center. Under H_0 ,

$$T^2 = n(\bar{X} - \mu_0)S^{-1}(\bar{X} - \mu_0)' \sim T^2(p, n-1) \text{ why?}$$

Since $\sqrt{n}(\bar{x} - \mu_0) \sim N(0, \Sigma)$, $(n-1)S \sim W_p(n-1, \Sigma)$, we have

$$T^2 = (n-1)[\sqrt{n}(\bar{x} - \mu_0)]'[(n-1)S]^{-1}[\sqrt{n}(\bar{x} - \mu_0)] \sim T^2(p, n-1)$$

Property: $\frac{n-p}{p(n-1)} T^2 \sim F(p, n-p).$

§4.3 Two Population Mean Comparative Inference

To master the construction of the Hotelling statistic in different scenarios and the corresponding test procedures:

- (1) Construct the Hotelling statistic for different scenarios based on its definition.
- (2) Determine the null distribution (F-distribution, with specified degrees of freedom) of the constructed Hotelling statistic using its properties.
- (3) Given the confidence level α , establish the rejection region and make a decision.

How to construct Hotelling statistics for the following tests

- i) Testing whether the mean vectors of two independent samples are the same (§4.3.1)

$$H_0: \mu_1 = \mu_2, \quad H_1: \mu_1 \neq \mu_2 \quad (4.3.1)$$

- ii) Testing the structural relationship of elements in a single population mean vector (§4.4)

$$H_0: C\mu = \varphi, \quad H_1: C\mu \neq \varphi \quad (4.4.2)$$

- iii) Testing the structural relationship of elements in two population mean vectors (§4.5)

$$H_0: C(\mu_1 - \mu_2) = \varphi, \quad H_1: C(\mu_1 - \mu_2) \neq \varphi$$

Examples in this chapter

§4.4 Contour analysis (轮廓分析)

Example 4.3.2.

§4.5 Multiple Population Mean Comparative Test

Under multivariate normal population assumption,

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k, H_1: \mu_i \neq \mu_j, \text{ at least one couple } i \neq j \quad (4.5.1)$$

Wilks Λ Statistic: $\Lambda_{p,k-1,n-k} = |SSE|/|SST|$,

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}})(\mathbf{x}_{ij} - \bar{\mathbf{x}})' \longrightarrow df = n - 1$$

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)' \longrightarrow E \longrightarrow df = n - k$$

Properties: For a given significance level α , the rejection rule is:

$$\text{If } \Lambda_{p,k-1,n-k} \leq \Lambda_{p,k-1,n-k,\alpha}, \text{ reject } H_0$$

where the critical value $\Lambda_{p,k-1,n-k,\alpha}$ satisfies: when H_0 is true,

$$P(\Lambda_{p,k-1,n-k} \leq \Lambda_{p,k-1,n-k,\alpha}) = \alpha$$

Example 4.6.

§4.7 The covariance matrix equality Test

Under multivariate normal population assumption,

Consider hypothesis test:

$$H_0: \Sigma_1 = \Sigma_2 = \dots = \Sigma_k, \quad H_1: \Sigma_i \neq \Sigma_j, \text{ at least one couple } i \neq j \quad (4.7.1)$$

Box-M test, How to derive Box-M statistic

It can be proved that (Exercise 1) likelihood ratio statistic

$$\lambda = \frac{|A|^{-n/2}}{\prod_{i=1}^k |A_i|^{-n_i/2}},$$

where $A_i = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)'(x_{ij} - \bar{x}_i)$ and $A = \sum_{i=1}^n \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)'(x_{ij} - \bar{x}_i)$ such that $A = A_1 + \dots + A_k$.

A modification of the above LRT statistic based on unbiasedness is

$$\lambda = \frac{|S_p|^{(n-k)/2}}{\prod_{i=1}^k |S_i|^{(n_i-1)/2}}, \quad (4.7.2)$$

where $S_i = \frac{1}{n_i-1} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)(x_{ij} - \bar{x}_i)'$, $\bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij}$, and $S_p = \frac{1}{n-k} \sum_{i=1}^k (n_i - 1) S_i$.

Then, Box' M statistic is defined as:

$$M = -2 \ln \lambda = (n - k) \ln |S_p| - \sum_{i=1}^k (n_i - 1) \ln |S_i|. \quad (4.7.3)$$

Chap 5: Discriminant Analysis,

§5.2 Distance Discriminant : Two group, multiple groups

- 1) Distance discrimination simplifies to linear discrimination for the case where the covariance matrices of the two sets of distances are the same;
- 2) Understanding the formula for distance discrimination under the assumption of a multivariate normal distribution.

§5.3 Bayesian Discriminant: maximum posterior probability criteria,

- Suppose $\pi_1, \pi_2, \dots, \pi_k$ are k groups and the probability density function of group π_i is $f_i(\mathbf{x})$. Prior probability of sample $\mathbf{x}$ from π_i is $p_i, i=1, 2, \dots, k$, satisfying $p_1 + p_2 + \dots + p_k = 1$. Then posterior probabilities of $\mathbf{x}$ belonging to π_i are

$$P(\pi_i | \mathbf{x}) = \frac{p_i f_i(\mathbf{x})}{\sum_{i=1}^k p_i f_i(\mathbf{x})}, \quad i = 1, 2, \dots, k$$

where $p_i = P(\pi_i)$, $f_i(\mathbf{x}) = P(\mathbf{x} | \pi_i)$.

- Maximum posterior probability criterion's discriminant rule:

$$\mathbf{x} \in \pi_l, \quad \text{if } P(\pi_l | \mathbf{x}) = \max_{1 \leq i \leq k} P(\pi_i | \mathbf{x}) \quad (5.3.2)$$

■ Minimum Expected Cost of Misclassification (ECM) criter

Two groups:

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases}$$

Multiple groups:

Minimum ECM discriminant rule is

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{\substack{j=1 \\ j \neq l}}^k p_j f_j(\mathbf{x}) c(l|j) = \min_{1 \leq i \leq k} \sum_{\substack{j=1 \\ j \neq i}}^k p_j f_j(\mathbf{x}) c(i|j) \quad (5.3.23)$$

Example 5.3.1-5.3.4

§5.4 Fisher Discriminant

- Understanding the basic steps of Fisher's discrimination through Example 5.4.2
- The Fisher discriminant method implies the assumption of multivariate normal distribution when finding the optimal projection direction to distinguish between groups

Example 5.4.2

Exercise 5.1. For Fisher discriminant, we need to assume $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k$. Suppose $\mathbf{x}_{ij}$ are p-dimensional observed value from π_i , $j=1,2,\dots,n_i$, $i=1,2,\dots,k$. Project them to a certain p-dimensional constant variable, a. The projection points will be $y_{ij} = \mathbf{a}'\mathbf{x}_{ij}$. Let the error sum of squares $SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 = \mathbf{a}'E\mathbf{a}$ and the total regression sum of squares $SSTR = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2 = \mathbf{a}'H\mathbf{a}$, where $E = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)(x_{ij} - \bar{x}_i)'$, $H = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})(\bar{x}_i - \bar{x})'$. In order to separate the mean values of each group after projection as far as possible, we should choose such kind of a so that the following F-value

$$F = \frac{SSTR}{SSE}$$

can attain the maximum, that is

$$\Delta(\mathbf{a}) = \frac{SSTR}{SSE} = \frac{\mathbf{a}'H\mathbf{a}}{\mathbf{a}'E\mathbf{a}}$$

attains the maximum. This results in the r nonzero eigenvectors $t_1, t_2, \dots, t_r$ of _____ with the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$ (2°) are the first r best directions we need. Therefore, the first r discriminants for Fisher discrimination are _____.

§5.5 *Stepwise discrimination

Chap 6: Cluster Analysis

§6.2 Distance and Similarity Coefficient, Shortest distance method,

- Master the use of the shortest distance method and the longest distance method to classify samples, and draw the dendrogram of the clustering process.

§6.3 6 methods for cluster analysis

- Master average linkage (类平均) method and understanding other methods

§ 6.3 Hierarchical Cluster (系统聚类)Method

- **Hierarchical Clustering Method** is the most used method among all the cluster analysis methods.
- Basic idea:
 - Step 1: treat each of n sample as a cluster and stipulate the sample-to-sample and cluster-to-cluster distances.
 - Step 2: combine the two clusters with the smallest distance into a new cluster and calculate the distance between the new cluster and others respectively;
 - Step 3: reduce a cluster each time by repeating step 2 until all the sample are in one cluster.
- 8 kind methods will be introduced. The difference is calculation method of the distance between clusters.

§6.4 Dynamic clustering method –understand K-means method.

Chap 7 Principal Component Analysis (PCA)

§7.2 (for population) PCA: definition, derivation, properties,

- Principal component analysis does not require the population to follow a normal distribution, which is different from Fisher discriminant method based on the assumption of population normal distribution

§7.3 Application of PCA (For samples) Sample PCA:

Example 7.3.1-7.3.3.

Chap 8 Factor Analysis

§8.2 Orthogonal factor analysis model

- Common factors are unobservable.
- The estimation of the factor loading matrix is not unique.
- Orthogonal factor models and their properties:
- The meaning of communality variance h_i^2 .
- The meaning of sum of squares of the column elements g_j^2 .

§8.3 Parameter estimation

Principal component analysis method (**Remember**) ,

S is approximately decomposed to : $S = \hat{A}\hat{T}\hat{A}', \hat{T} = (\hat{t}_1, \dots, \hat{t}_p)$

$$\begin{aligned} S &= \hat{\lambda}_1 \hat{t}_1 \hat{t}_1' + \dots + \hat{\lambda}_m \hat{t}_m \hat{t}_m' + \hat{\lambda}_{m+1} \hat{t}_{m+1} \hat{t}_{m+1}' + \dots + \hat{\lambda}_p \hat{t}_p \hat{t}_p' \\ &\approx \hat{\lambda}_1 \hat{t}_1 \hat{t}_1' + \dots + \hat{\lambda}_m \hat{t}_m \hat{t}_m' + \hat{D} = \hat{A}\hat{A}' + \hat{D} \end{aligned} \quad (8.3.1)$$

where $\hat{A} = (\sqrt{\hat{\lambda}_1} \hat{t}_1, \dots, \sqrt{\hat{\lambda}_m} \hat{t}_m) = (\hat{a}_{ij})$ is $p \times m$ matrix, $\hat{D} = \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_p^2)$, $\hat{\sigma}_i^2 = s_{ii} - \sum_{j=1}^m \hat{a}_{ij}^2$, $i=1, 2, \dots, p$. $\hat{A}$ and $\hat{D}$ is one **principal component solution** (主成分分解) of factor model.

Factorization method (principal factor solution Remember),

ML method

Both PCA method and Factorization method are based on such a

Decomposition: $\Sigma = AA' + D$ or $R = AA' + D$,

where A is used for the estimation of the factor loading matrix,
and D is used for the estimation of variance of special factors.

Example 8.3.1-8.3.2, Exercises

§8.4 Factor rotation: The purpose of factor rotation is to achieve
distinct factor loadings, which allows for a reasonable interpretation of
the common factors in the model and reduces subjectivity.

Example 8.4.1-8.4.2

§8.5 Factor score

Two common methods for estimating factor scores include:

1, Weighted least square estimation (WLSE):

$$\hat{f} = (A'D^{-1}A)^{-1}A'D^{-1}(x_j - \mu) \quad (8.5.3) \text{ (for population)}$$

$$\hat{f} = (\hat{A}'\hat{D}^{-1}\hat{A})^{-1}\hat{A}'\hat{D}^{-1}(x_j - \bar{x}) \quad (8.5.4) \text{ (for samples)}$$

If the original variables have already been standardized.

$$\hat{f} = (\hat{A}'\hat{D}^{-1}\hat{A})^{-1}\hat{A}'\hat{D}^{-1}x^*, \text{ where } \hat{A} = ?, \hat{D} = ?$$

2, Regression methods:

$$\tilde{f} = E(f|x) = A'\Sigma^{-1}(x - \mu) \quad (8.5.8) \text{ (For population)}$$

$$\tilde{f} = \hat{A}'S^{-1}(x - \bar{x}) \quad (8.5.11) \text{ (For samples)}$$

If the original variables have already been standardized:

$$\tilde{f} = \hat{A}^*'\hat{R}^{-1}x^*, \text{ see Example 8.5.1}$$

It is not necessary to remember the above formulas.

Chap 10 Canonical Correlation Analysis

§ 10.2 Population Canonical Correlations

The definitions of population canonical variables and population canonical correlation coefficients

► Suppose $\mathbf{x}=(x_1, x_2, \dots, x_p)'$ and $\mathbf{y}=(y_1, y_2, \dots, y_q)'$ are two random variables, $\text{Cov}(\mathbf{x})=\Sigma_{11}(>0)$, $\text{Cov}(\mathbf{y})=\Sigma_{22}(>0)$, $\text{Cov}(\mathbf{x}, \mathbf{y})=\Sigma_{12}$. Then we have:

$$\text{Cov}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

where $\Sigma_{21}=\Sigma_{12}'$.

We will study the correlation between $u=\mathbf{a}'\mathbf{x}$ and $v=\mathbf{b}'\mathbf{y}$, where $\mathbf{a}=(a_1, a_2, \dots, a_p)'$, $\mathbf{b}=(b_1, b_2, \dots, b_q)'$

$$\rho(u, v) = \frac{\mathbf{a}'\Sigma_{12}\mathbf{b}}{\sqrt{\mathbf{a}'\Sigma_{11}\mathbf{a}}\sqrt{\mathbf{b}'\Sigma_{22}\mathbf{b}}} \quad (10.2.3)$$

► Under the constraints:

$$V(u)=1, \quad V(v)=1 \quad (10.2.4)$$

that is

$$\mathbf{a}'\Sigma_{11}\mathbf{a}=1, \quad \mathbf{b}'\Sigma_{22}\mathbf{b}=1 \quad (10.2.5)$$

we hope to find some $\mathbf{a} \in R^p$ some $\mathbf{b} \in R^q$ such that

$$\rho(u, v)=\mathbf{a}'\Sigma_{12}\mathbf{b} \quad (10.2.6)$$

achieves the maximum.

To solve the optimization problem in (10.2.5)- (10.2.6), let $\beta_1, \beta_2, \dots, \beta_m$

be the m positive orthonormal eigenvectors of

$\Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}$ corresponding to eigenvalues $\rho_1^2, \rho_2^2, \dots, \rho_m^2$.

Denote by $\alpha_i = \Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}\beta_i/\rho_i$, $a_i = \Sigma_{11}^{-1/2}\alpha_i$, $b_i = \Sigma_{22}^{-1/2}\beta_i$.

Then, the following conclusions holds.

Proposition 1 (Exer 1):

- $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ are orthonormal eigenvectors of $\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2} \Sigma_{21}$ corresponding to $\rho_1^2, \rho_2^2, \dots, \rho_m^2$.
- $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ are eigenvectors of $\Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ corresponding to $\rho_1^2, \rho_2^2, \dots, \rho_m^2$.
- $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_m$ are eigenvectors of $\Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$ corresponding to $\rho_1^2, \rho_2^2, \dots, \rho_m^2$.

Under Proposition 1, we can prove that when $\mathbf{a} = \mathbf{a}_1 = \Sigma_{11}^{-1/2} \alpha_1$, $\mathbf{b} = \mathbf{b}_1 = \Sigma_{22}^{-1/2} \beta_1$, $\rho(u, v) = \mathbf{a}' \Sigma_{12} \mathbf{b}$ attains the maximum $\rho_1 = \alpha_1' \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2} \beta_1$. So, we take

$u_1 = \mathbf{a}_1' \mathbf{x}, \quad v_1 = \mathbf{b}_1' \mathbf{y}$
as the first pair of canonical variables, denote $\mathbf{a}_1, \mathbf{b}_1$ as the first pair of canonical coefficient vectors, denote ρ_1 as the first canonical correlation.

10.3 Sample Canonical Correlations

- The definitions of sample canonical variables and sample canonical correlation coefficients, and how to find them

Example 10.3.1

10.4 Testing the significance of the canonical correlations

Example 10.4.1 and Example 10.4.2